

Referee Report: Resolving Coordination Frictions in Green Labor Transitions: Minimizing Unemployment, Costs, and Welfare Distortions (EEREV-D-25-01160)

This paper develops a two-sector search and matching model to study labor reallocation into green jobs. Of particular interest is the role of policy in addressing the two-sided coordination failure that arises between workers and firms. The model extends the standard DMP model in several ways. First, while workers are free to search for non-green jobs, switching to the green sector is costly. Firms are free to post vacancies in either sector, but the government is assumed to subsidize (tax) productive matches in the green (non-green) sector. The government can also provide support to workers who switch into the green sector by subsidizing their switching cost.

The authors use their model to consider the U.S. energy transition, which is expected to increase green energy employment from 2% to 14% by 2030. Comparing steady states, the authors show that each policy alone is capable of reaching the 14% employment target, but the costs associated with each policy varies. As such, a combined policy that incentivizes both worker and firm participation is found to outperform the individual policies. The authors also consider the dynamics of the energy transition using a sequence of steady states. This exercise provides further insights into the state-contingent nature of the optimal policy, with public support focusing on workers early and late in the transition, and firm support being more dominant near the middle.

While I believe that this paper considers an important and timely question, I have the following concerns regarding the paper in its present form.

1 Comments

1. The conclusion that coordinated policy support on both sides of the labor market dominates one-sided interventions seems to follow naturally from the model's DMP structure. In such environments, searching workers and firms create externalities, so incentivizing each side of the market has a reinforcing effect in the labor market. For this reason, the directional result that dual-subsidies outperform worker-only or firm-only policies is not, by itself, surprising. As such, I do not see this initial steady state result as the primary contribution. Instead, the paper's value lies in the mapping of the optimal policy along during the green transition (more on this in subsequent

comments). The authors should revise their language to better focus on their primary contribution.

2. Given that the transition is modeled as a sequence of policy-induced steady states, not as a true dynamic transition with forward-looking agents, the authors should rephrase claims about “optimal dynamic policy” more cautiously (e.g., “optimal policy along steady-state targets”).
3. Building on the previous point, while the authors state that “this transition study is deliberately not a full perfect-foresight transition,” it is not clear that steady state conditions would be met at each of these intermediate steps. As such, analysis that accounts for the full perfect-foresight transition, while also meeting intermediate goals/restrictions, is preferred. Linking to my first point, if the majority of the value added of this paper is related to policy dynamics, it is necessary to consider the complete transition dynamics for comparison purposes.
4. The authors assume linear utility when conducting welfare analysis. The authors should add a table varying risk aversion or at least discuss how curvature in the utility function would affect the CEV thresholds.